Resource-Constrained Echocardiogram Video Classification: 3D CNNs vs. Vision Transformers Under a 6 GB VRAM Constraint

Abstract

Heart failure affects roughly 64 million people worldwide. About half of these cases are heart failure with reduced ejection fraction (HFrEF), which leads to frequent hospitalizations and high mortality. Automatically detecting reduced EF from echocardiogram videos could make diagnoses faster and more consistent than manual checks. However, the newest 3D CNNs and vision transformers usually require high-end hardware.

In this project, we compared a 3D CNN (R(2+1)D-18, ~31.5M parameters) and a vision transformer (MVIT v2 Small, ~34.5M parameters) on the EchoNet-Dynamic dataset, keeping both strictly under a 6 GB VRAM constraint. Both models were pretrained on Kinetics-400. We tested them using two methods: (1) linear probing (training a logistic regression model on frozen features) and (2) transfer learning (fine-tuning the models end-to-end).

During linear probing, the models performed similarly. MVIT had a slightly higher ROC-AUC (0.828 vs 0.804), but the difference was not statistically significant. After fine-tuning, both models improved significantly, but the MVIT pulled ahead (ROC-AUC 0.936 vs 0.902; sensitivity at ~90% specificity 0.825 vs 0.725). Bootstrap analysis confirmed this advantage was statistically significant. Our results show that while both models can be trained on consumer GPUs, the MViT achieved better performance in the fine-tuned setting under the 6 GB VRAM constraint.

1. Introduction

Heart failure is a rapidly growing public health crisis that currently affects an estimated 64 million people worldwide [1]. Roughly 50% of these cases are classified as heart failure with reduced ejection fraction (HFrEF), a severe condition defined by a left ventricular ejection fraction (EF) of 40% or less [2]. HFrEF is associated with frequent hospitalizations, poor quality of life, and high mortality rates [1]. While echocardiography is the gold standard for assessing EF, manual tracing of ventricular volumes is conventionally time-consuming and suffers from high inter-operator variability [3]. The automated detection of reduced EF using deep video models has the potential to improve both the speed and consistency of these critical clinical

assessments [4].

However, VRAM is a major barrier for many modern 3D/video and 3D medical imaging models, with common configurations spanning roughly 9–40 GB depending on architecture and setup. Demonstrating a resource-efficient pipeline on a 6 GB RTX 3050 would therefore be an important accessibility result, even if it does not generalize to all SOTA models without model-specific optimizations.

To build a highly efficient pipeline, this project directly compares two distinct architectural families with similar parameter counts, holding pretraining data (Kinetics-400) [5] constant see how their underlying designs perform on ultrasound data:

- **3D Convolutional Neural Networks (CNNs):** We utilize the R(2+1)D-18 architecture [6] as our baseline. CNNs possess a strong local inductive bias and translation invariance, making them historically well-suited for processing the noisy, speckle-heavy textures inherent to ultrasound imaging.
- **Vision Transformers (ViTs):** We challenge the baseline with the Multiscale Vision Transformer (MVIT v2 Small) [7]. Unlike CNNs, Transformers utilize a global self-attention mechanism, theoretically allowing them to better model the long-range, non-linear temporal dynamics of the cardiac cycle across frames.

This project was a hands-on way to learn about PyTorch, transfer learning, and handling memory constraints. It also served as a practical exercise in building robust data pipelines, maintaining data hygiene, and applying rigorous, reproducible evaluation methods in machine learning. Our main question was: Can Vision Transformers beat 3D CNNs on noisy medical videos, and can we actually train them with only 6 GB of VRAM?

2. Methodology

2.1 Dataset & Preprocessing

- **Data Source:** We used the EchoNet-Dynamic dataset [8], comprising 10030 echocardiogram videos. We define Heart Failure with reduced Ejection Fraction (HFrEF) using the clinical threshold[2], mapping reduced ejection fraction (EF ≤ 40) to a positive label .
- **Quality Control:** Automated probes verify that videos can be successfully decoded at multiple parts of each video (start, middle, and end) rather than assuming validity from the file presence alone. We utilize the given splits (75/12.5/12.5 training/validation/test) and verify that each video file is unique to prevent test set leakage.
- **Video Sampling:** Videos are processed by extracting 16-frame clips with a temporal stride of 2 frames to capture a sufficiently large range of cardiac motion. We reference the methodology applied in EchoNet-Dynamic, which used 32-frame clips with stride 2 [8]. During training, clips are sampled from a random temporal starting point, whereas

validation and testing utilize a deterministic, centered temporal window.

2.2 Model Architectures

This study aims to emphasize architectural inductive bias as the main comparison axis, while holding hardware and pretraining data (Kinetics-400) [5] constant.

- **Baseline (R(2+1)D-18):** We utilize a 3D Convolutional Neural Network [6], [9] with ~31.5M parameters. This architecture relies on local temporal processing via spatiotemporal convolutions.
- **Challenger (MVIT v2 Small):** We compare the CNN against a Multiscale Vision Transformer [7], [9] with ~34.5M parameters, which models global temporal dependencies through self-attention.
- **Linear Probing:** Model heads were dynamically modified based on the training phase. For Phase 1 feature extraction, the original classification heads were removed to directly produce a 512-dimensional continuous embedding for the CNN and a 768-dimensional embedding for the Transformer, which are used to train a logistic classifier.
- **Fine Tuning:** The R(2+1)D head was replaced with a single Linear layer, while the MVIT head was replaced with a Sequential block containing Dropout (p=0.5) and a Linear layer.

2.3 Training Pipeline (Two-Phase Approach)

- **Phase 1 (Frozen Feature Extraction & Linear Probing):**
  To strictly test the quality of representations transferred from natural video, we froze the backbone parameters and extracted the resulting features. A Z-score normalization (StandardScaler) [10], fitted strictly on the training set, was applied to the extracted embeddings before training a deterministic Scikit-Learn Logistic Regression model using the L-BFGS solver [11]. We chose to use a deterministic algorithm with the intent of truly capturing the base state of each model.
- **Phase 2 (End-to-End Fine-Tuning):**
  We selectively unfroze the deepest layers of both architectures. For the R(2+1)D model, we unfroze the final spatiotemporal block (layer4), and for the MVIT v2, we unfroze the final transformer blocks alongside the layer normalization (norm) parameters. Models were optimized using AdamW [12], [13] with a ReduceLROnPlateau scheduler. During final evaluation, we implemented clip-averaging, with reference to the EchoNet-Dynamic study (sampling 5 random clips per video and averaging their predicted probabilities) [8] to ensure the full cardiac cycle was represented.

2.4 Hardware Constraints & Efficient Training

- **Compute Bottleneck:** Training was conducted locally on an NVIDIA GeForce RTX 3050 Laptop GPU with exactly 6GB of VRAM.
- **Strategic Unfreezing:** To fit training into 6GB of VRAM, we employed strategic unfreezing. By keeping the majority of the early backbone layers frozen, PyTorch is not required to store gradients for those parameters in the computational graph. This drastically reduces the memory overhead required during the backward pass, preventing Out-Of-Memory

(OOM) errors.

- **Mixed Precision (AMP):** To further minimize the memory footprint and accelerate training, we utilized PyTorch Automatic Mixed Precision (AMP) [14] with a GradScaler. This dynamically casts operations to 16-bit floats (FP16) where mathematically safe, rather than relying on standard 32-bit floats, keeping VRAM usage strictly within the 6GB hardware limit.

3. Evaluation Metrics & Statistical Analysis

3.1 Primary and Secondary Performance Metrics

To comprehensively compare the models, we utilized a combination of several metrics:

- **ROC-AUC:** Serving as our primary metric, the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) evaluates how well the models discriminate between normal ejection fraction and HFrEF across all classification thresholds.
- **F1-Score and Accuracy:** To assess point-performance and class imbalance handling, we report the F1-score and overall accuracy. To strictly prevent data leakage and overly optimistic reporting, the optimal probability threshold that maximizes the F1-score is evaluated exclusively on the validation set. This threshold is then permanently locked before evaluating the predictions on the test set.

3.2 Clinical Operating Point

In clinical deployment, false positives for heart failure can lead to unnecessary, expensive, and often stressful downstream testing. Therefore, we define a strict clinical operating target:

- **Sensitivity at 90% Specificity:** We require the models to operate at a strict 90% specificity target. Similar to the F1-score, the exact probability threshold required to achieve a False Positive Rate (FPR) of 0.10 is determined using the validation set's ROC curve. This locked threshold is applied to the test set to report the realized sensitivity and specificity under simulated clinical constraints.

3.3 Statistical Significance Testing

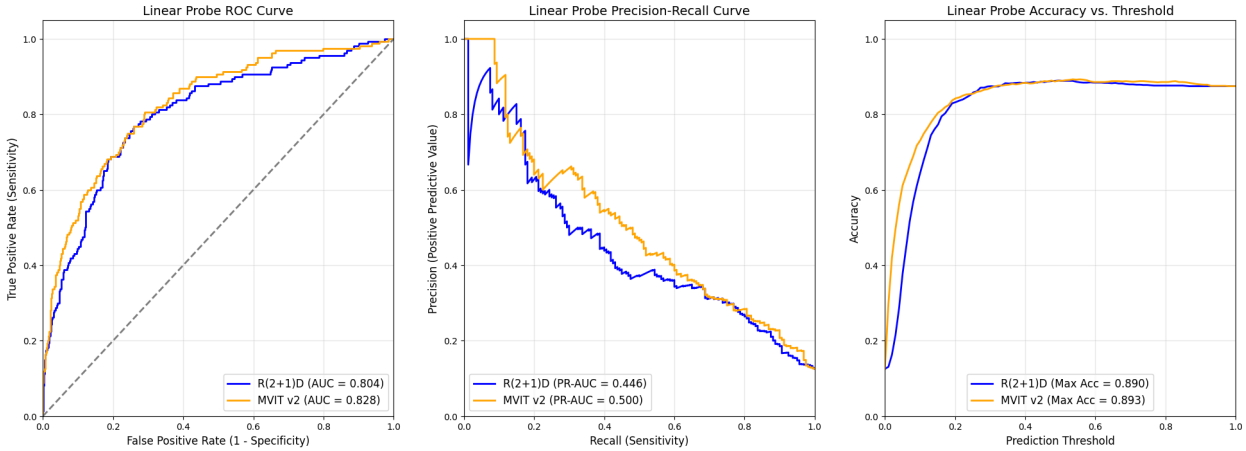
To rigorously determine whether the performance difference between the global self-attention of the MVIT and the local convolutions of the R(2+1)D is statistically significant, we employ a non-parametric approach:

- **Paired Bootstrap Analysis:** We utilize a paired bootstrap test with n=1000 iterations.
- **Methodology:** Because both models evaluate the exact same test dataset, we resample the test-set indices with replacement 1000 times using a fixed random seed to ensure reproducibility. For each bootstrap sample, we calculate the difference in AUC between the two architectures (Delta AUC = MViT AUC - R(2+1)D AUC).
- **Confidence Intervals:** We compute a 95% Confidence Interval (CI) by extracting the 2.5th and 97.5th percentiles of the resulting Delta AUC distribution. If this 95% CI does not encompass zero, we can reject the null hypothesis and we infer a statistically significant

ROC-AUC difference on this test set under our evaluation protocol.

4. Results

4.1 Phase 1: Linear Probe Performance



The linear probe tests how well the models perform using only the features they learned from everyday YouTube videos of the Kinetics-400 dataset. In this constrained setting, the MVIT v2 demonstrated a slight edge over the R(2+1)D across all primary metrics on the test set. As shown in Table 1, the Transformer achieved higher overall discriminatory ability and better sensitivity at our strict clinical operating target.

However, statistical testing revealed that this numerical advantage is not conclusive. A paired bootstrap analysis of the test-set ROC-AUCs yielded a mean difference of 0.0243 in favor of the MVIT. Because the 95% confidence interval for this difference [-0.0097, 0.0615] encompasses zero, the linear-probe comparison is ultimately a statistical tie. This indicates that when relying solely on Kinetics-400 pretraining, neither the global self-attention of the Transformer nor the local convolutions of the CNN strictly dominate on cardiac ultrasound video.

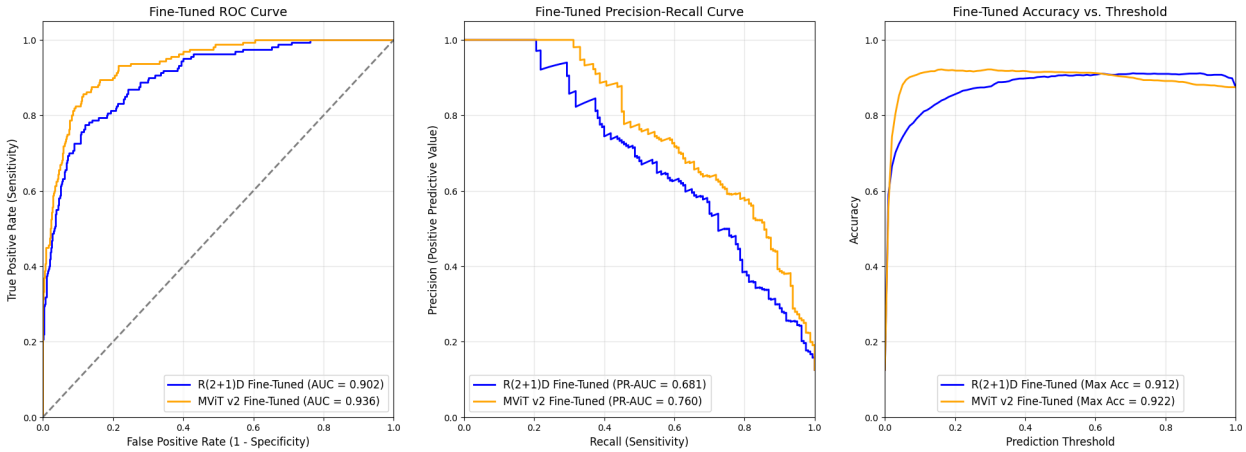
Table 1: Phase 1 (Linear Probing) Results

Metric	R(2+1)D-18	MVIT v2 Small
ROC-AUC	0.8039	0.8280
PR-AUC	0.446	0.500
Sensitivity (at ~90% Specificity)	0.4625	0.5188
F1-Score	0.4099	0.4685

Accuracy	0.8692	0.8810

(Note: PR-AUC values are derived from the generated precision-recall curves. Thresholds were locked on the validation set to prevent leakage. For the F1-optimized operating point, locked thresholds were 0.3859 for MVIT and 0.2825 for R(2+1)D. For the ~90% specificity target, locked thresholds were 0.2560 for MVIT and 0.2183 for R(2+1)D, resulting in realized specificities of 0.9051 and 0.8926, respectively.)

4.2 Phase 2: Fine-Tuning & Clip-Averaging Performance



After we allowed the models to fine-tune on the actual echocardiograms and used clip-averaging, both models improved drastically. In this phase, the MViT showed higher performance across the reported threshold-agnostic metrics. As detailed in Table 2, the MVIT v2 outpaced the 3D CNN across our primary threshold-agnostic metrics.

At our defined clinical operating target of approximately 90% specificity, the MVIT v2 achieved a substantially higher sensitivity (0.8250) compared to the R(2+1)D model (0.7250). The ROC and PR curves are visually consistent with this trend, which is supported by the paired bootstrap analysis. The mean ROC-AUC difference between the models was 0.0343, with a 95% confidence interval of [0.0132, 0.0567]. Because this interval lies entirely above zero,we conclude that the MViT had significantly higher ROC-AUC than R(2+1)D in the fine-tuned EchoNet-Dynamic test setting.

Table 2: Phase 2 (Fine-Tuning) Results

Metric	R(2+1)D-18	MVIT v2 Small
ROC-AUC	0.9020	0.9359
PR-AUC	0.6806	0.7596
Sensitivity (at ~90% Specificity)	0.7250	0.8250
Max Accuracy	~0.912	~0.922

(Note: Default threshold (t=0.5) metrics yielded an F1 of 0.6203 and accuracy of 0.9060 for the CNN, compared to an F1 of 0.4883 and accuracy of 0.9146 for the MVIT. Clinical operating thresholds were locked at 0.3234 and 0.0628, respectively.)

4.3 Computational Efficiency and Training Dynamics

While the Transformer had better accuracy, it was much harder to train. Even though we unfroze fewer parameters for the MVIT (~13.6M) than the CNN (~23.5M), the MVIT's self-attention math used a lot more memory.

- **Memory Constraints and Batch Sizing:** To prevent Out-Of-Memory (OOM) errors on the 6GB VRAM constraint, the MVIT required halving the batch size relative to the CNN. This reduction doubled the number of optimization steps per epoch (1867 iterations for MVIT versus 934 for R(2+1)D).
- **Training Throughput:** The R(2+1)D architecture maintained a training throughput of roughly 5.7 iterations per second, completing a full training epoch in approximately 2 minutes and 45 seconds. In contrast, the MVIT processed batches at roughly 3.1 iterations per second, requiring nearly 10 minutes per training epoch—an approximate 3.6x increase in training time.
- **Inference Speed:** During the final test-set evaluation using 5-clip averaging, the CNN processed the 1277 test videos at a rate of 3.29 iterations per second (completing in 6 minutes and 28 seconds). The MVIT processed the same test set at 1.43 iterations per second (completing in 14 minutes and 55 seconds).

Table 3: Computational Profile

Metric	Baseline: R(2+1)D-18	Challenger: MVIT v2 Small
Trainable Parameters	23,496,037	13,615,489
Steps per Epoch (Train)	934	1867
Avg. Train Time / Epoch	~2 min 45 sec	~9 min 54 sec
Avg. Val Time / Epoch	~50 sec	~2 min 09 sec
Total Test Inference Time	6 min 28 sec	14 min 55 sec
Best Validation AUC	0.9216 (Epoch 9)	0.9370 (Epoch 9)

5. Discussion & Conclusion

5.1 Interpretation

Our results highlight the differences between local CNNs and global Transformers. Before fine-tuning (Phase 1), the models tied. This suggests that the shared Kinetics-400 pretraining likely acted as a performance ceiling for both models. Because Kinetics-400 [5] consists of natural, everyday videos, the extracted embeddings were not optimized for the noisy, speckle-heavy textures of echocardiograms.The generic pretraining was a bottleneck for both of them.

After fine-tuning on echocardiogram data, the MViT achieved higher ROC-AUC and higher sensitivity at the predefined ~90% specificity operating point than the R(2+1)D model. This pattern was supported by paired bootstrap analysis of ROC-AUC. These findings suggest that end-to-end adaptation was important for cross-domain transfer from Kinetics-400 to echocardiography in our setup. The MViT advantage is consistent with the hypothesis that global attention may better capture long-range temporal structure, although this mechanism was not directly tested here.

The downside is the hardware cost. Despite the MVIT exposing roughly 10 million fewer trainable parameters than the R(2+1)D-18 during fine-tuning, its self-attention matrix caused significant memory overhead, requiring the batch size to be halved to prevent Out-Of-Memory errors. Even though the Vision Transformer won in accuracy, this meant that we had to train it with smaller batch sizes, making it nearly four times slower to train than the CNN. In resource-limited settings where frequent retraining is needed, the CNN may still be the more practical choice despite the slight drop in accuracy.

5.2 Limitations & Future Work

Because of our 6GB memory limit, we trained on short 16-frame clips. A 16-frame clip doesn't always capture a full heartbeat, which is why we averaged 5 different clips together during testing, to simulate full-cycle assessment. In the future, we could attempt to increase frame window and compensate by decreasing batch sizes.

We hypothesize that domain-specific pretraining could yield larger gains than architecture changes alone, but this remains to be tested. Recently, a wave of foundation models built specifically for echocardiography has been released, including models like EchoJEPA [16], EchoCLIP [17], and EchoFM [15]. These models are pretrained large amounts of medical images rather than general YouTube videos.

Our logical next step is to replace the Kinetics-400 weights with weights from these domain-specific models. However, a critical part of this future work will be testing whether these newer, potentially larger foundation models can still be efficiently fine-tuned under our strict 6 GB VRAM constraint, or if they will require more aggressive memory-saving techniques like gradient checkpointing or lower-rank adaptations (LoRA) to remain viable on consumer hardware.

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Appendix: Data Availability and Use Restrictions

Dataset. This study uses the EchoNet-Dynamic echocardiogram video dataset (Stanford University School of Medicine), as described in the Methodology section.

Access and permitted use. EchoNet-Dynamic is distributed under a Research Use Agreement

(RUA). Access is restricted to registered users and is limited to personal, non-commercial research use. Users must register individually with Stanford to obtain access.

Use restrictions relevant to this project. In accordance with the EchoNet-Dynamic RUA, we do not redistribute any portion of the dataset, do not share dataset download links, and do not provide copies of the original data files. To comply with the EchoNet-Dynamic data use restrictions, this repository/paper does not include:

- raw echocardiogram videos,
- redistributed dataset copies,
- direct dataset download links,
- pretrained/fine-tuned weights derived from EchoNet-Dynamic.

Code availability. Code for preprocessing, training, and evaluation is provided here:

https://github.com/Gandalfthegloom/ecg-hfref-classifier

Reproducibility note. Researchers who wish to reproduce these results should independently register for EchoNet-Dynamic access through Stanford and run the provided code using the same train/validation/test split protocol and preprocessing settings described.